WEEKLY TEST -2

GIT BRANCHES

- Using “git branch <name>” command to create an new branch “git branch” command to check branch status “git checkout <branchname>” command to switch branch.

```
MINGW64:/c/Users/hariharan/Downloads/giteve6/haridevops
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 285 bytes | 142.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/hharan1899-lgtm/haridevops.git
   4db9c5f..a79ae6a  main -> main

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (main)
$ git branch hari/newbranch

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (main)
$ git branch
  hari/newbranch
* main

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (main)
$ git checkout hari/newbranch
Switched to branch 'hari/newbranch'

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (hari/newbranch)
$ git branch
* hari/newbranch
  main

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (hari/newbranch)
$ git add .

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (hari/newbranch)
$ nano file7

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (hari/newbranch)
$ git add .
warning: in the working copy of 'file7', LF will be replaced by CRLF the next time Git touches it

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/Downloads/giteve6/haridevops (hari/newbranch)
$ git commit -m "fifthcommit"
[hari/newbranch f1262dc] fifthcommit
 1 file changed, 2 insertions(+)
 create mode 100644 file7
```

GIT REMOTE

- Using “git remote -v” command to check repository connect with their local repository to be fetch and push it.

```
MINGW64:/c/Users/hariharan/downloads/haridevops

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~
$ mkdir giteve6
mkdir: cannot create directory 'giteve6': File exists

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~
$ mkdir giteve6
mkdir: cannot create directory 'giteve6': File exists

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~
$ cd downloads

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads
$ mkdir giteve6
mkdir: cannot create directory 'giteve6': File exists

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads
$ cd haridevops/
bash: cd: haridevops/: No such file or directory

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads
$ git clone https://github.com/hharan1899-lgtm/haridevops.git
Cloning into 'haridevops'...
remote: Enumerating objects: 39, done.
remote: Counting objects: 100% (39/39), done.
remote: Compressing objects: 100% (21/21), done.
remote: Total 39 (delta 11), reused 27 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (39/39), 4.16 KiB | 157.00 KiB/s, done.
Resolving deltas: 100% (11/11), done.

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads
$ cd haridevops/

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/haridevops (main)
$ git remote -v
origin  https://github.com/hharan1899-lgtm/haridevops.git (fetch)
origin  https://github.com/hharan1899-lgtm/haridevops.git (push)

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/haridevops (main)
$
```

GIT PULL

- Using “git pull origin <branchname>” command to pull remote repository to local repository.

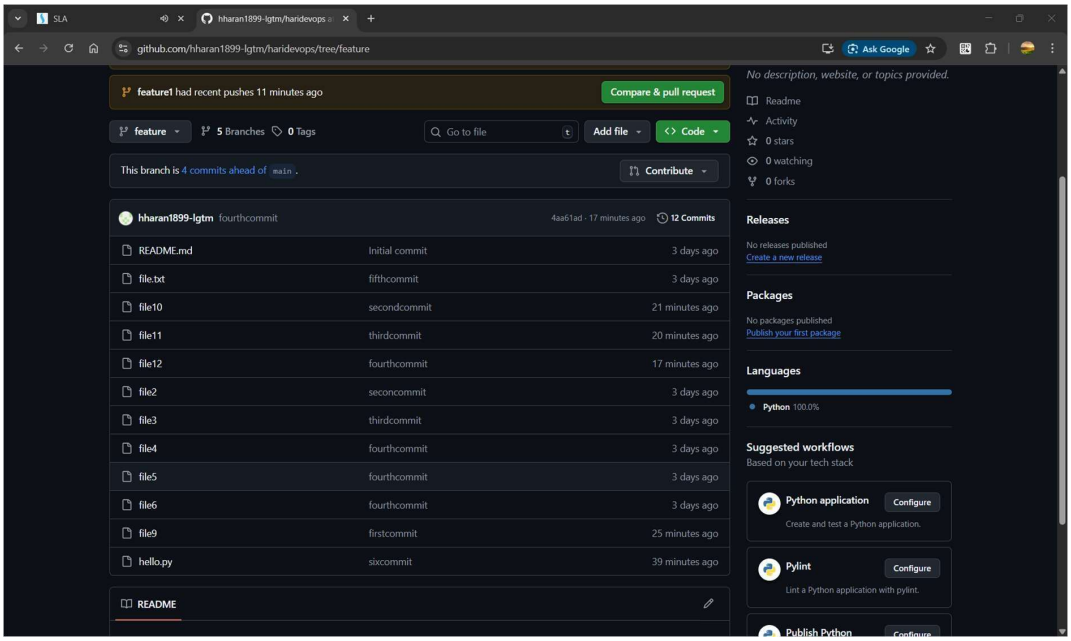
```
MINGW64:/c/Users/HARI/downloads/gitrev/todayeve

HARI@LAPTOP-60HJK66L MINGW64 ~/downloads/gitrev/todayeve (feature1)
$ git commit -m second try"
> ^C

HARI@LAPTOP-60HJK66L MINGW64 ~/downloads/gitrev/todayeve (feature1)
$ git commit -m "second try"
[feature1 6139823] second try
 1 file changed, 1 insertion(+)
 create mode 100644 index.py

HARI@LAPTOP-60HJK66L MINGW64 ~/downloads/gitrev/todayeve (feature1)
$ git push origin feature1
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 290 bytes | 290.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Hari-haran1788/todayeve.git
   0d5e42e..6139823  feature1 -> feature1

HARI@LAPTOP-60HJK66L MINGW64 ~/downloads/gitrev/todayeve (feature1)
$ git pull origin feature2
From https://github.com/Hari-haran1788/todayeve
 * branch            feature2   -> FETCH_HEAD
Already up to date.

HARI@LAPTOP-60HJK66L MINGW64 ~/downloads/gitrev/todayeve (feature1)
$ git switch feature2
```

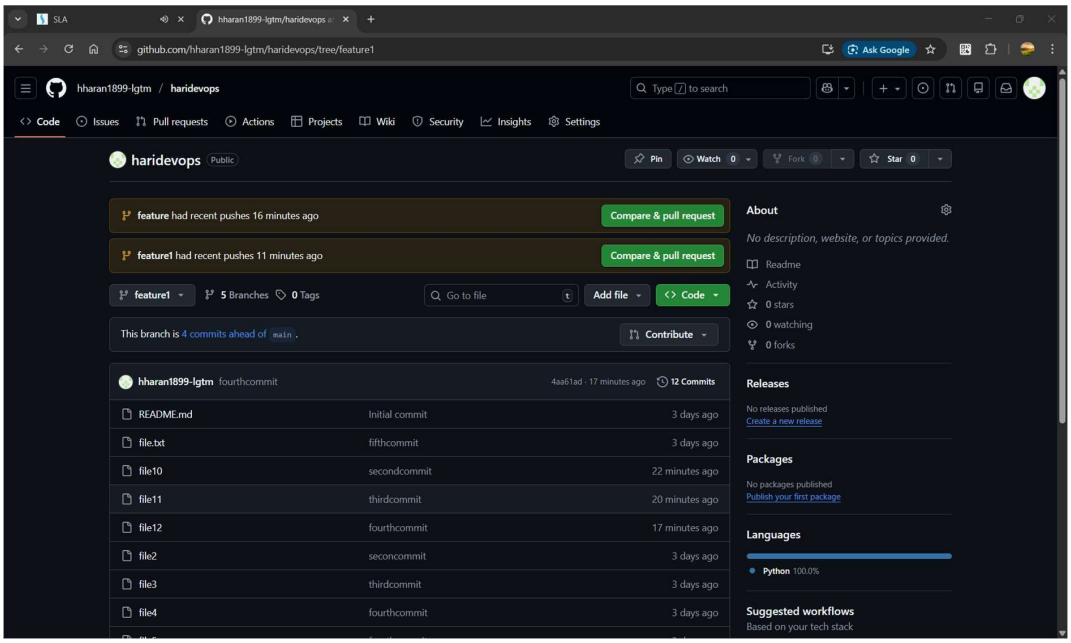
GIT MERGE

- Using “git checkout feature1” command to switch the branch “git merge feature” command to merge an two branch.

```
MINGW64:/c/Users/hariharan/giteve6/haridevops
$ git commit -m "fourthcommit"
[feature 4aa61ad] fourthcommit
 1 file changed, 1 insertion(+)
 create mode 100644 file12

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature)
$ git push origin feature
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 272 bytes | 136.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/hharan1899-lgtm/haridevops.git
   71ba3db..4aa61ad  feature -> feature

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature)
$ git log --oneline
4aa61ad (HEAD -> feature, origin/feature) fourthcommit
71ba3db thirdcommit
0e278b7 secondcommit
f815beb firstcommit
401656a (origin/subbranch, origin/main, origin/HEAD, subbranch, main, feature1) sixcommit
568be7e Create hello.py
a79ae6a fifthcommit
4db9c5f fourthcommit
cd6293f thirdcommit
b5588aa seconcommit
015e4ec firstcommit
2221d3c Initial commit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature)
$ git checkout feature1
Switched to branch 'feature1'

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature1)
$ git log --oneline
401656a (HEAD -> feature1, origin/subbranch, origin/main, origin/HEAD, subbranch, main) sixcommit
568be7e Create hello.py
a79ae6a fifthcommit
```

```
MINGW64:/c/Users/hariharan/giteve6/haridevops
4db9c5f fourthcommit
cd6293f thirdcommit
b5588aa seconcommit
015e4ec firstcommit
2221d3c Initial commit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature)
$ git checkout feature1
Switched to branch 'feature1'

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature1)
$ git log --oneline
401656a (HEAD -> feature1, origin/subbranch, origin/main, origin/HEAD, subbranch, main) sixcommit
568be7e Create hello.py
a79ae6a fifthcommit
4db9c5f fourthcommit
cd6293f thirdcommit
b5588aa seconcommit
015e4ec firstcommit
2221d3c Initial commit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature1)
$ git merge feature
Updating 401656a..4aa61ad
Fast-forward
 file10 | 1 +
 file11 | 1 +
 file12 | 1 +
 file9  | 1 +
 4 files changed, 4 insertions(+)
 create mode 100644 file10
 create mode 100644 file11
 create mode 100644 file12
 create mode 100644 file9

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/giteve6/haridevops (feature1)
$ git log --oneline
4aa61ad (HEAD -> feature1, origin/feature, feature) fourthcommit
71ba3db thirdcommit
0e278b7 secondcommit
f815beb firstcommit
```





GIT REBASE

- Using “git rebase feature3” command to rewrite their commit history.

```
RAHUL - LOQ@RAHUL_LOQ MINGW64 ~/github/devops1 (feature1)
$ git rebase feature3
Current branch feature1 is up to date.
```

GIT INIT

- Using “git init” command used to create an new repository in a project folder. “.git” hidden files to be created.

```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame
$ cd downloads
bash: cd: downloads: No such file or directory

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame
$ git init
Initialized empty Git repository in C:/Users/hariharan/Downloads/harigame/.git/

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (master)
$ ls -la
total 44
drwxr-xr-x 1 hariharan 197121 0 Jan  6 18:43 ./
drwxr-xr-x 1 hariharan 197121 0 Jan  6 18:33 ../
drwxr-xr-x 1 hariharan 197121 0 Jan  6 18:43 .git/
```

- Using “git branch -m master main” command used to switch the branch master to main.

```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (master)
$ git branch -m master main

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git push origin main
```

GIT REMOTE

- Using “git remote -v” command does not list in repository using git init using “git remote add origin <link>” to list in remote repository.

```
$ git remote -v

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (master)
$ git remote add origin https://github.com/hharan1899-lgtm/harigame.git

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (master)
$ git remote -v
origin  https://github.com/hharan1899-lgtm/harigame.git (fetch)
origin  https://github.com/hharan1899-lgtm/harigame.git (push)
```

- Using “git push -f” command used to forceful push to an github.

```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git push -f origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 413 bytes | 137.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/hharan1899-lgtm/harigame.git
 + 2c3c947...bba58ae main -> main (forced update)

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
```

GIT RESET

- Using “git reset –soft <7 digit>” command used to head is move to an given file.

```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git reset --soft 4d91076

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   file4


hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git log --oneline
4d91076 (HEAD -> main) thirdcommit
bba58ae secondcommit
91f174a firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
```

- Using “git reset –mixed commit hash” command used to unstage of an file.

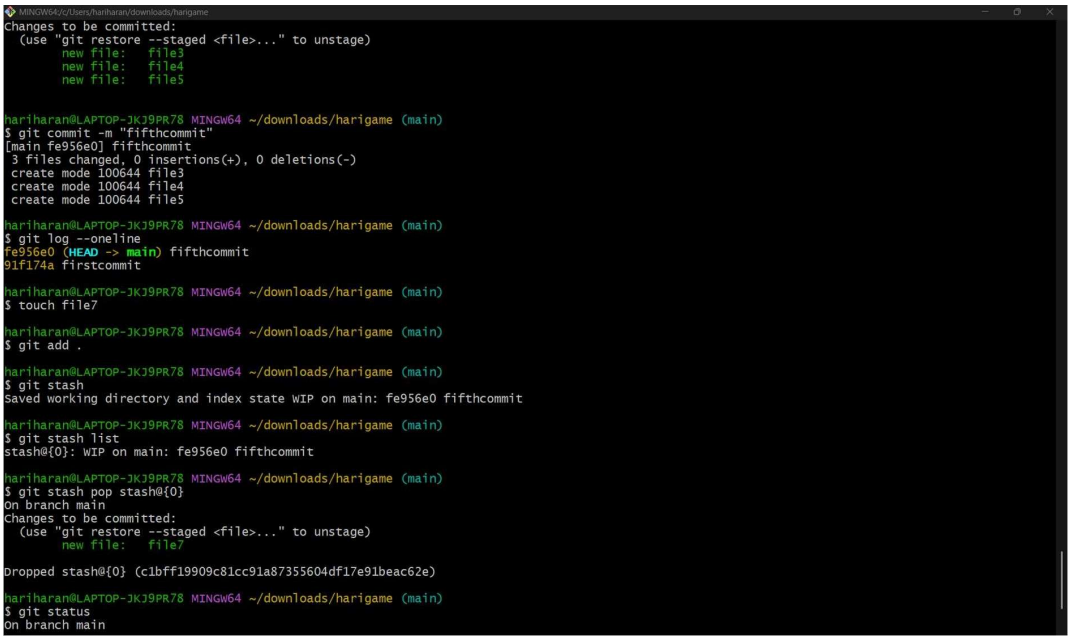
```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git reset --mixed commit bba58ae
fatal: ambiguous argument 'commit': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   file4


hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git reset --mixed bba58ae

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file3
        file4

nothing added to commit but untracked files present (use "git add" to track)

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git log --oneline
bba58ae (HEAD -> main) secondcommit
91f174a firstcommit
```

- Using “git reset –hard commit hash” command used to delete of this file.

```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git reset --hard 91f174a
HEAD is now at 91f174a firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        file3
        file4

nothing added to commit but untracked files present (use "git add" to track)

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git log --oneline
91f174a (HEAD -> main) firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ ls
file.txt  file3  file4
```

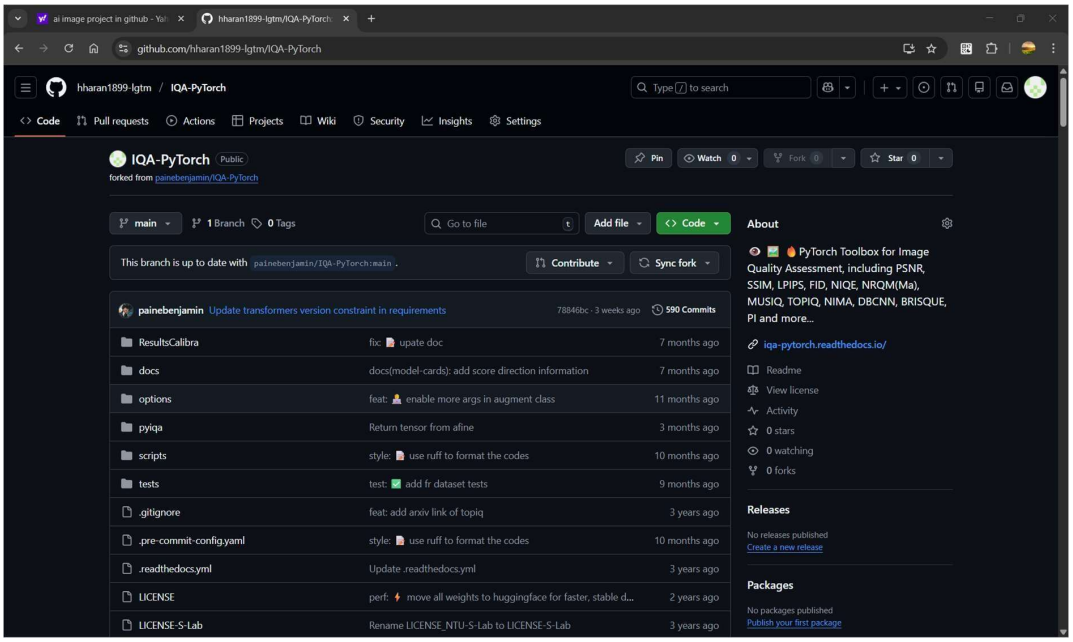
GIT STASH

- Using “git statsh” command used to save an temporary uncommit changes “git stash list” command to list an temporary uncommit changes “git stash pop” command to delete an uncommit changes.



GIT FORK

- Using git fork is a personal,server side copy of an entire repository under your own account allowing to freely experiment and making changes without affecting the original project.

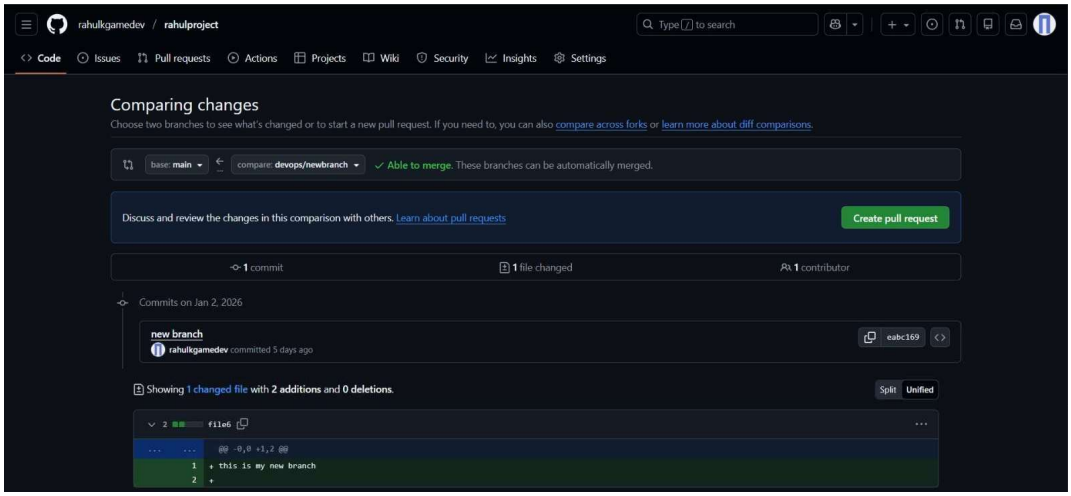


GIT PULL REQUEST

- A Pull Request (PR) is a way to request changes you made in one branch to be reviewed and merged into another branch (usually the main branch).

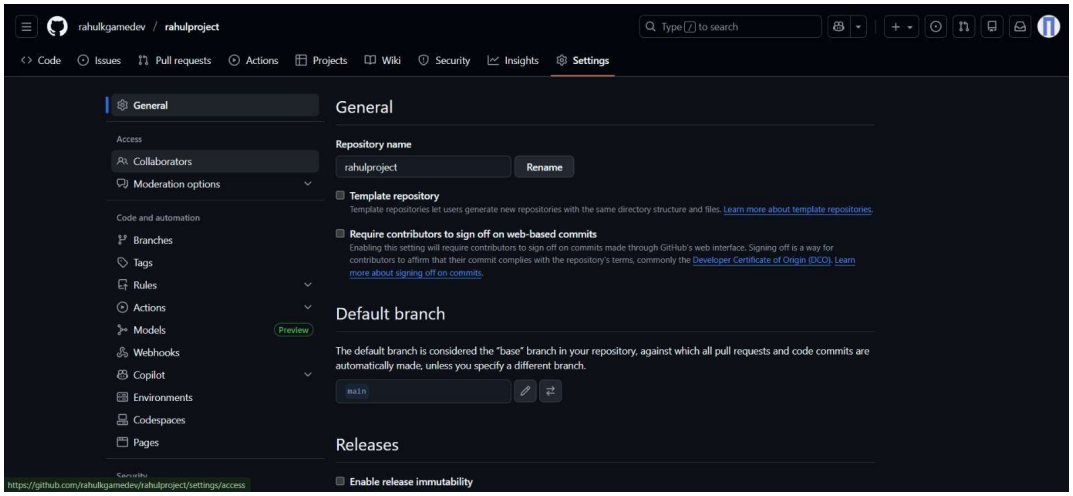
STEP 1:

Create a pull request



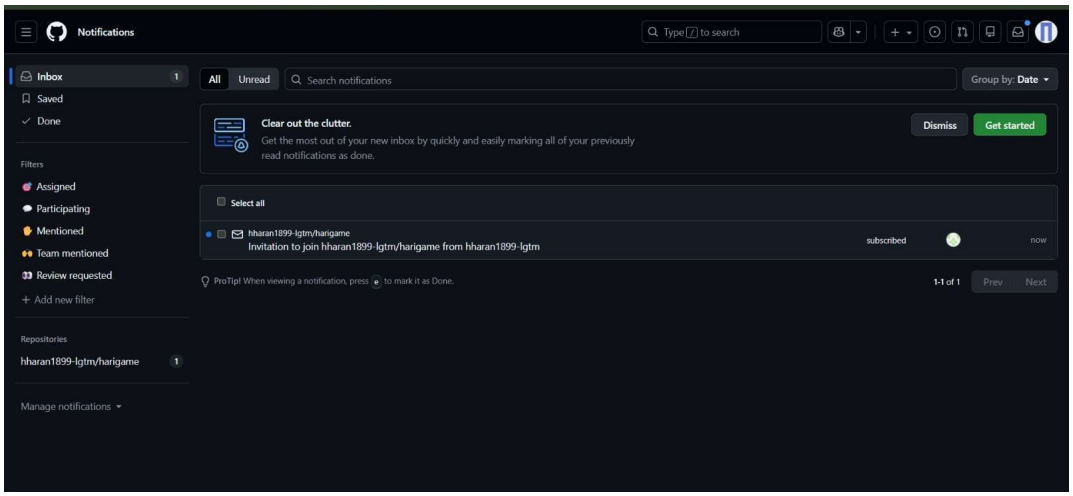
STEP 2:

To collaborates with the user



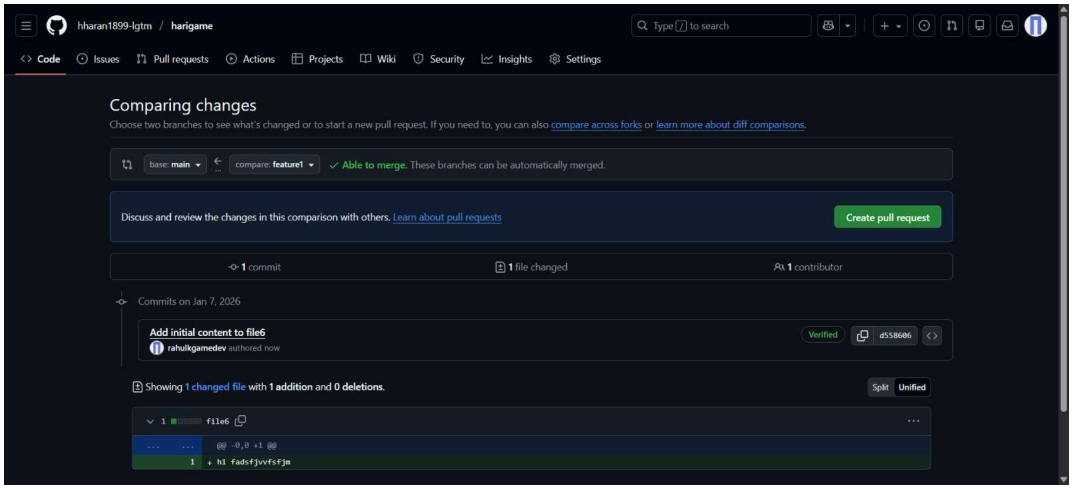
STEP 3:

To accept the invitation link



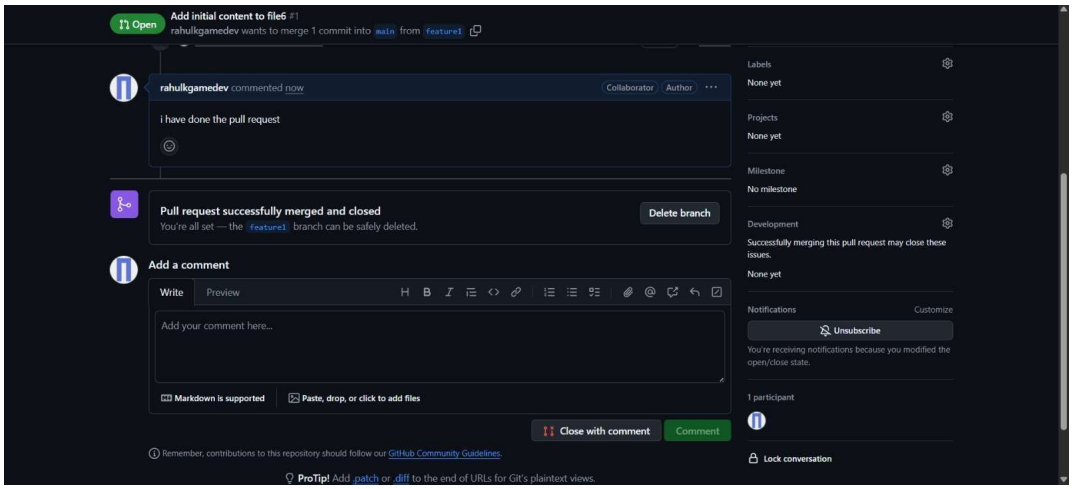
STEP:4

To assign the reviewer of their pull request file.



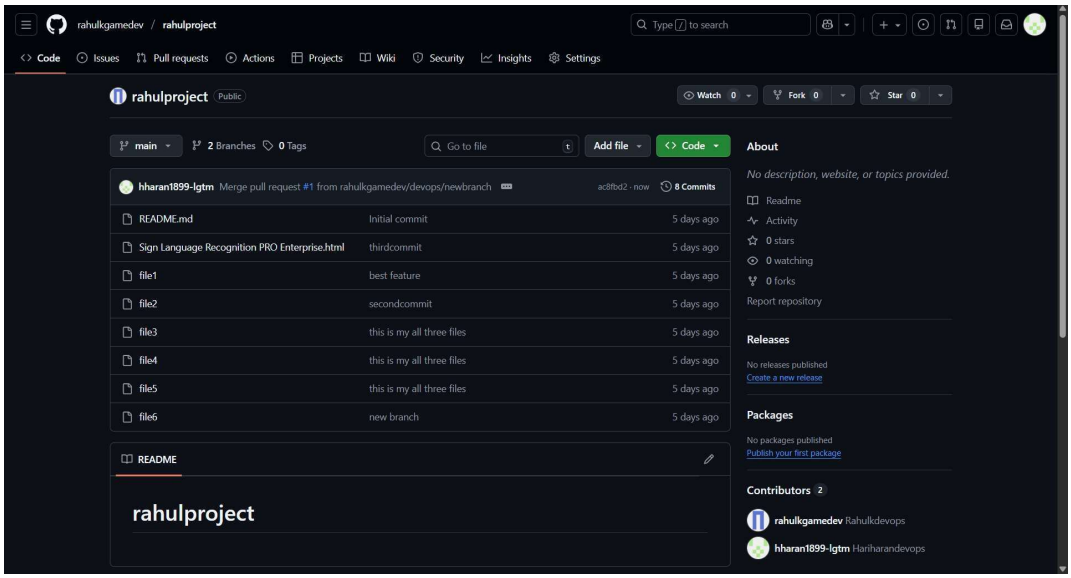
STEP:5

To review the file to tell suggest comment of an pull request file.



STEP:6

To merge pull request of their main branch.



GIT CHERRY PICK

Git is a git command to apply the changes introduced by one or more specific existing commits from one branch onto the current branch unlike command like merge or rebase,which integrate entire branches and their histories cherry pick offer a surgical approach to incorporating only the changes you need.

Using “git checkout <branch name>” command to switch the branch name “git cherry-pick <commit hash>” command allow you to pick specific commit existing commit from one branch onto the current branch.

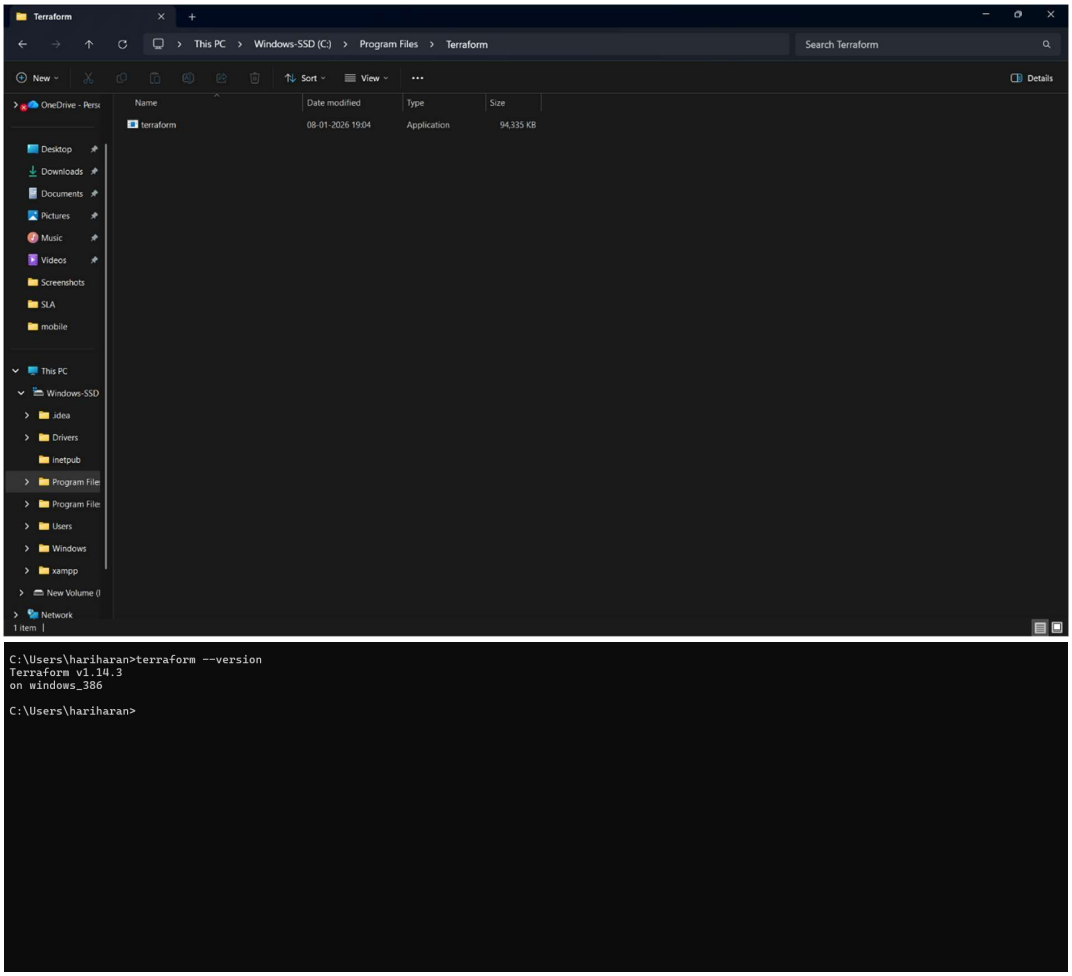
```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (feature2|CHERRY-PICKING)
$ git log --oneline
1ad82a5 (HEAD -> feature2) ninethcommit
9d3417a (feature1) seventhcommit
fe956e0 (main) fifthcommit
91f174a firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (feature2|CHERRY-PICKING)
$ git checkout main
Switched to branch 'main'
warning: cancelling a cherry picking in progress

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git log --oneline
fe956e0 (HEAD -> main) fifthcommit
91f174a firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git cherry-pick 9d3417a
[main e0607e0] seventhcommit
 Date: Wed Jan 7 18:55:06 2026 +0530
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file7
 create mode 100644 file8

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git log --oneline
e0607e0 (HEAD -> main) seventhcommit
fe956e0 fifthcommit
91f174a firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$
```

Using “git reflog –oneline” command to list the commited changes with commit id.

```
hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
$ git reflog --oneline
e0607e0 (HEAD -> main) HEAD@{0}: cherry-pick: seventhcommit
fe956e0 HEAD@{1}: checkout: moving from feature2 to main
1ad82a5 (feature2) HEAD@{2}: commit: ninethcommit
9d3417a (feature1) HEAD@{3}: checkout: moving from feature1 to feature2
9d3417a (feature1) HEAD@{4}: commit: seventhcommit
fe956e0 HEAD@{5}: checkout: moving from main to feature1
fe956e0 HEAD@{6}: reset: moving to HEAD
fe956e0 HEAD@{7}: commit: fifthcommit
91f174a HEAD@{8}: reset: moving to 91f174a
bba58ae HEAD@{9}: reset: moving to bba58ae
4d91076 HEAD@{10}: reset: moving to 4d91076
e3ee10b (origin/main) HEAD@{11}: commit: fourthcommit
4d91076 HEAD@{12}: commit: thirdcommit
bba58ae HEAD@{13}: commit: secondcommit
91f174a HEAD@{14}: Branch: renamed refs/heads/master to refs/heads/main
91f174a HEAD@{16}: commit (initial): firstcommit

hariharan@LAPTOP-JKJ9PR78 MINGW64 ~/downloads/harigame (main)
```

INSTALLATION OF TERRAFORM:

To install an terraform an terraform.io website to click install an latest version of an terraform.

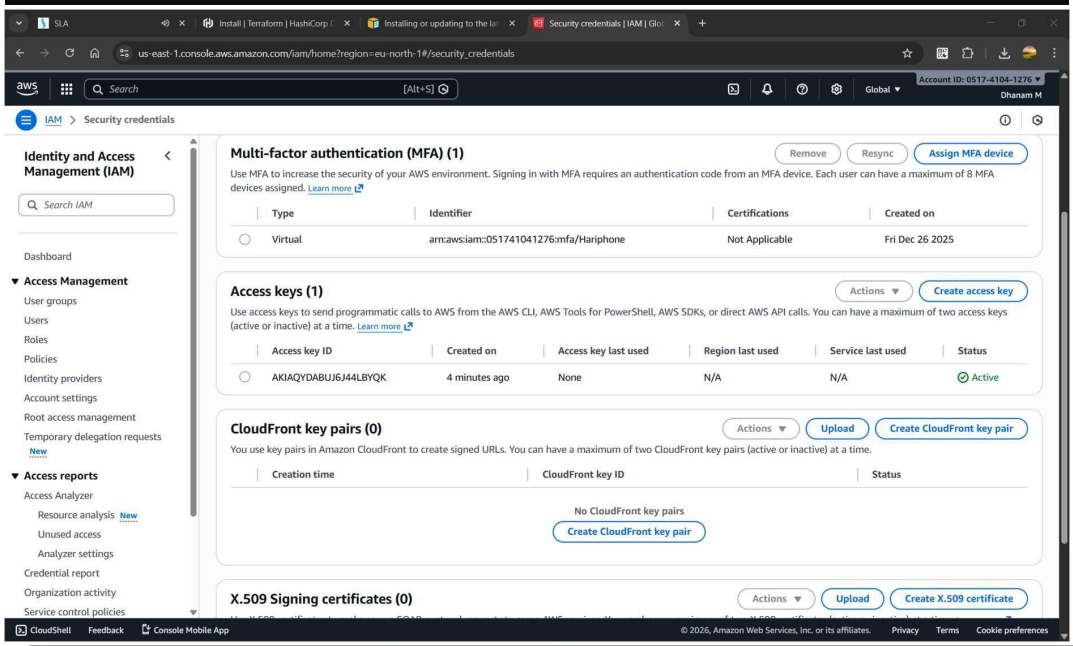


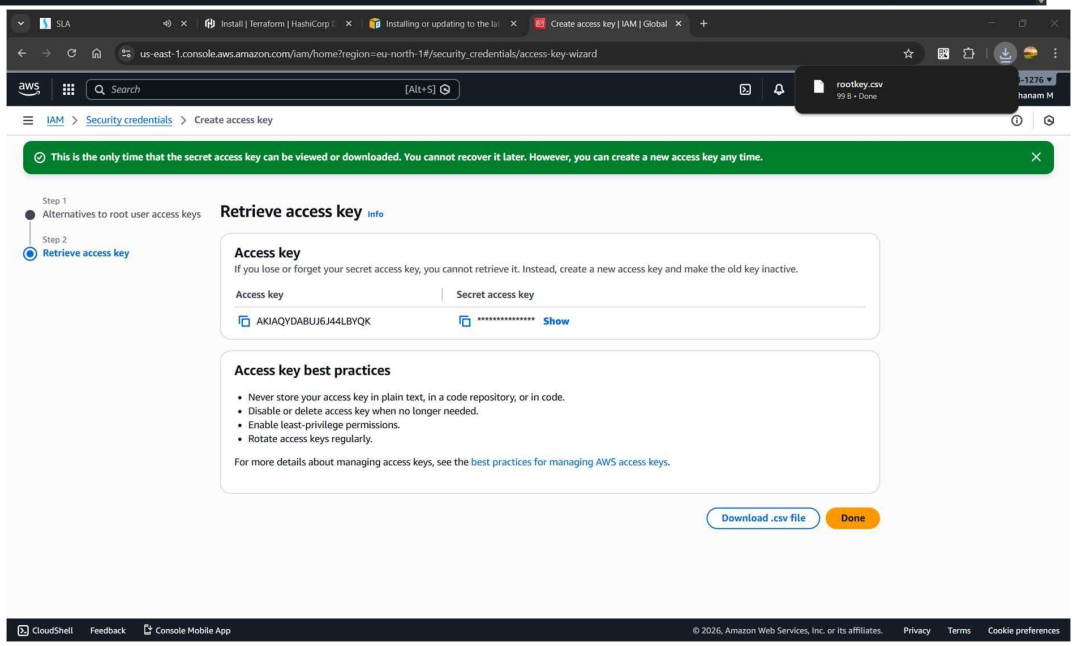
INSTALLATION OF AWS CLI:

To search an aws cli install in google chrome to click an website to install an latest version an aws cli.

To login an CLI need an access key and secret acess key of an aws account. Create an access key and secret access key.

```
C:\Users\hariharan>aws configure
AWS Access Key ID [None]:
AWS Secret Access Key [None]:
```

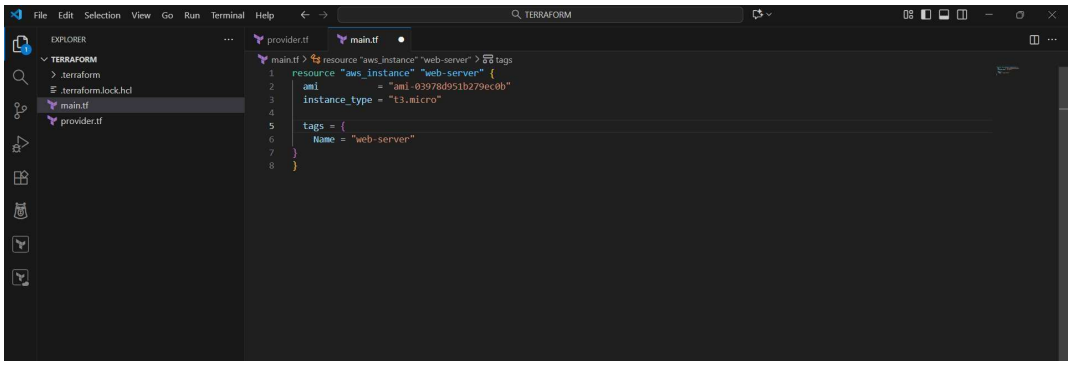


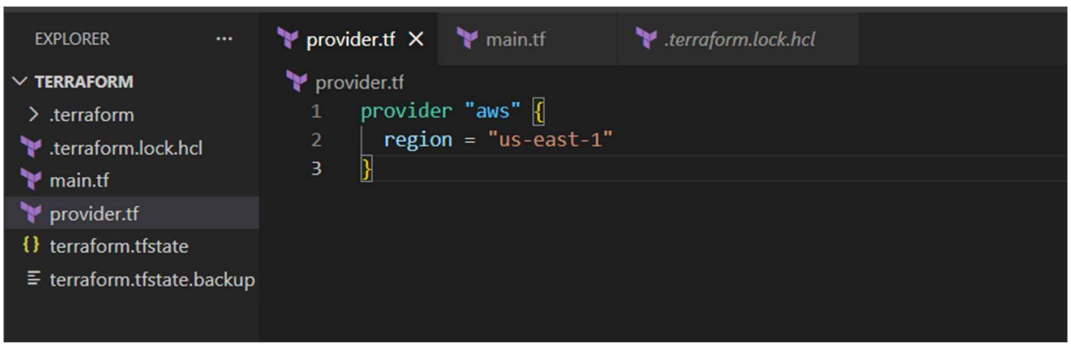


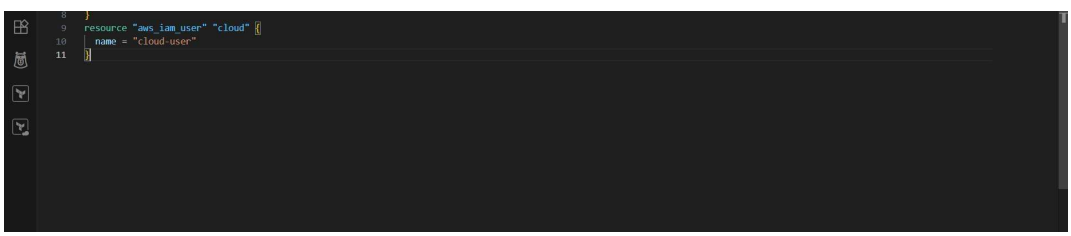
CREATE THE EC2 AND IAM

STEP:1

To open an visual studio open an new folder to save an file “.tf” to form terraform file to enter the code and run an terminal.

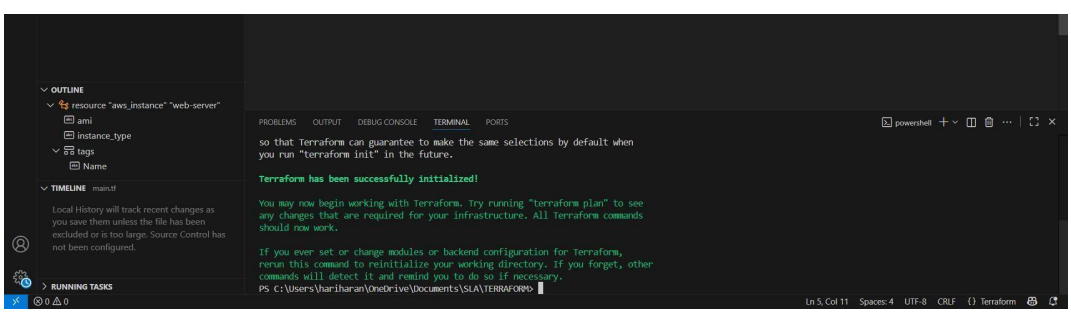






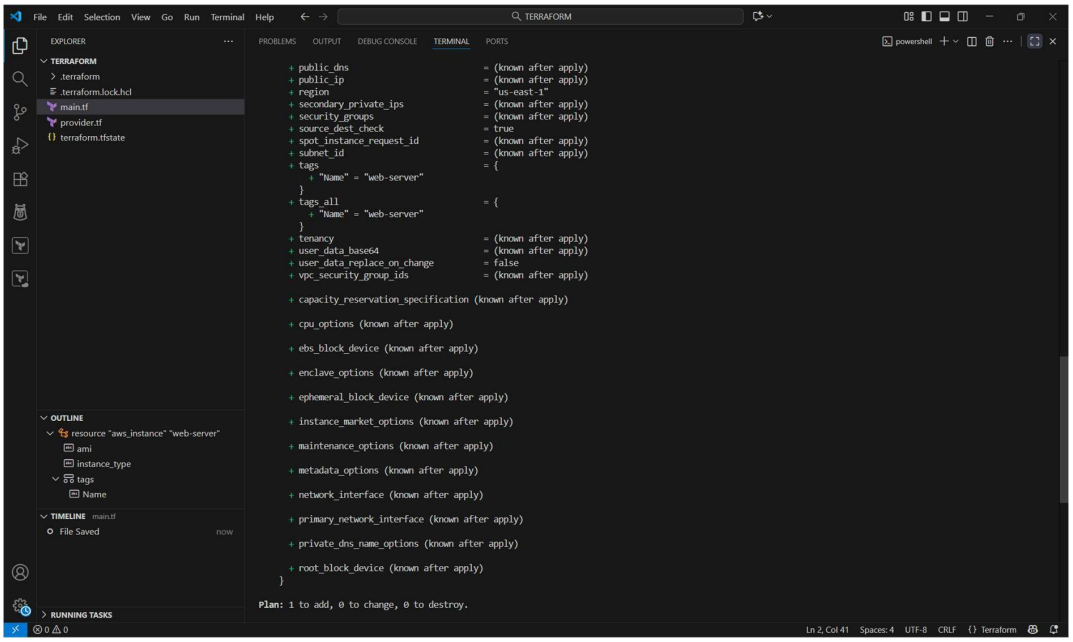
STEP:2

“TERRAFORM INIT” COMMAND to use an initialization of an given code.



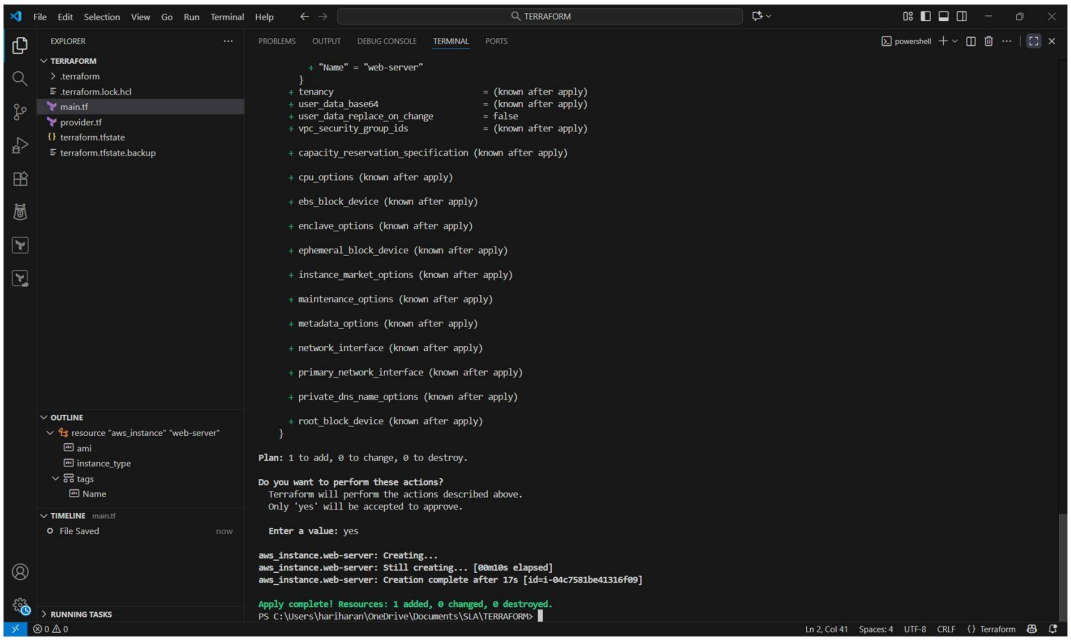
STEP:3

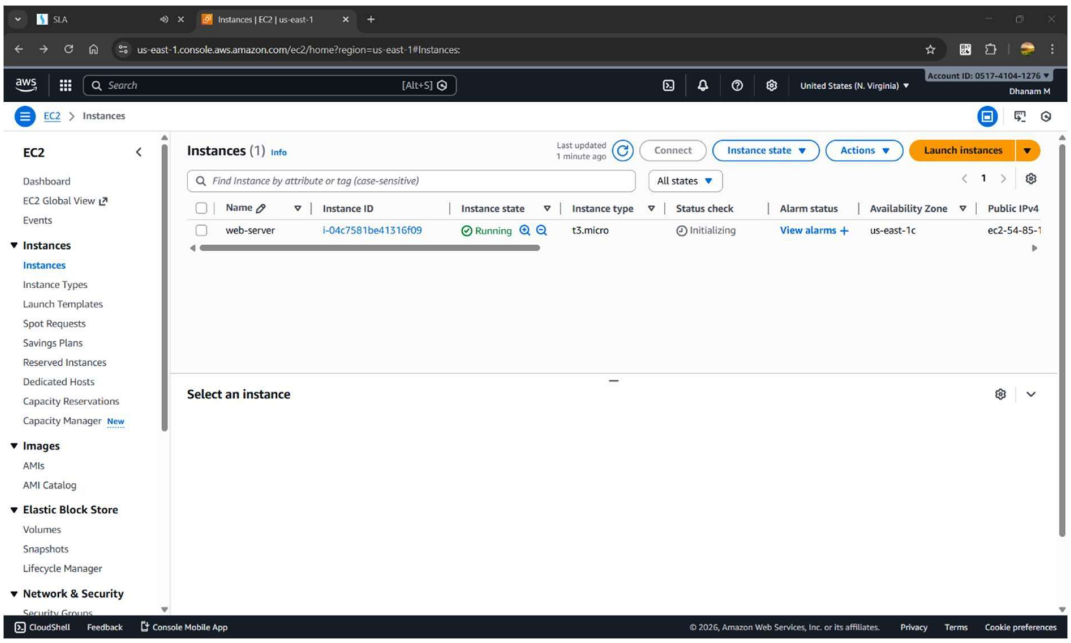
“TERRAFORM PLAN COMMAND” to use an plan of an given code.

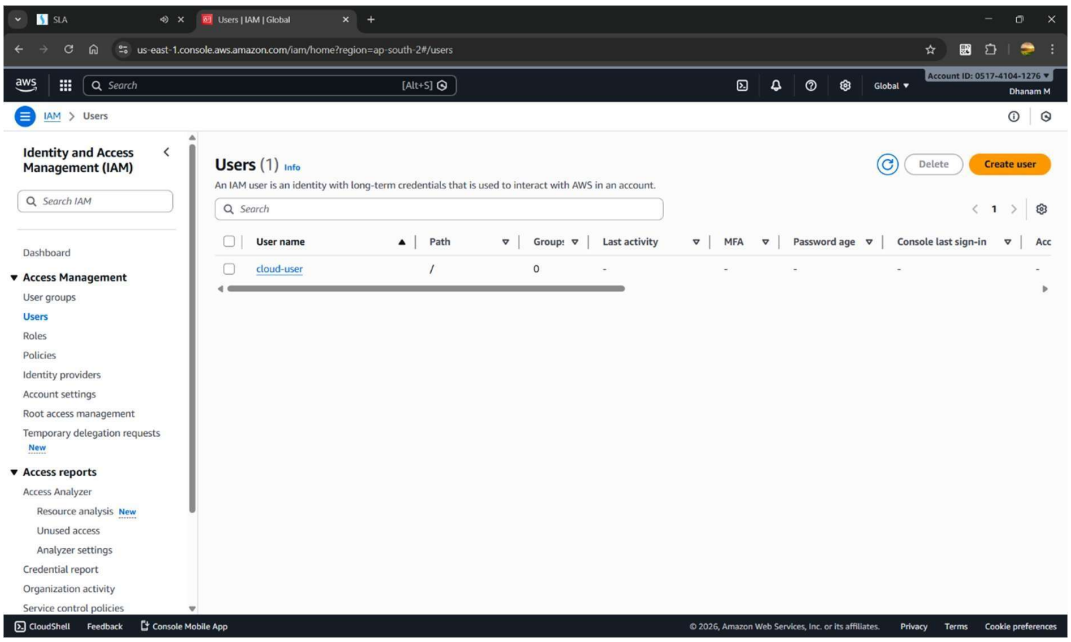


STEP:4

“TERRAFORM APPLY” COMMAND to use an apply to launch an instances in aws account for this given code.

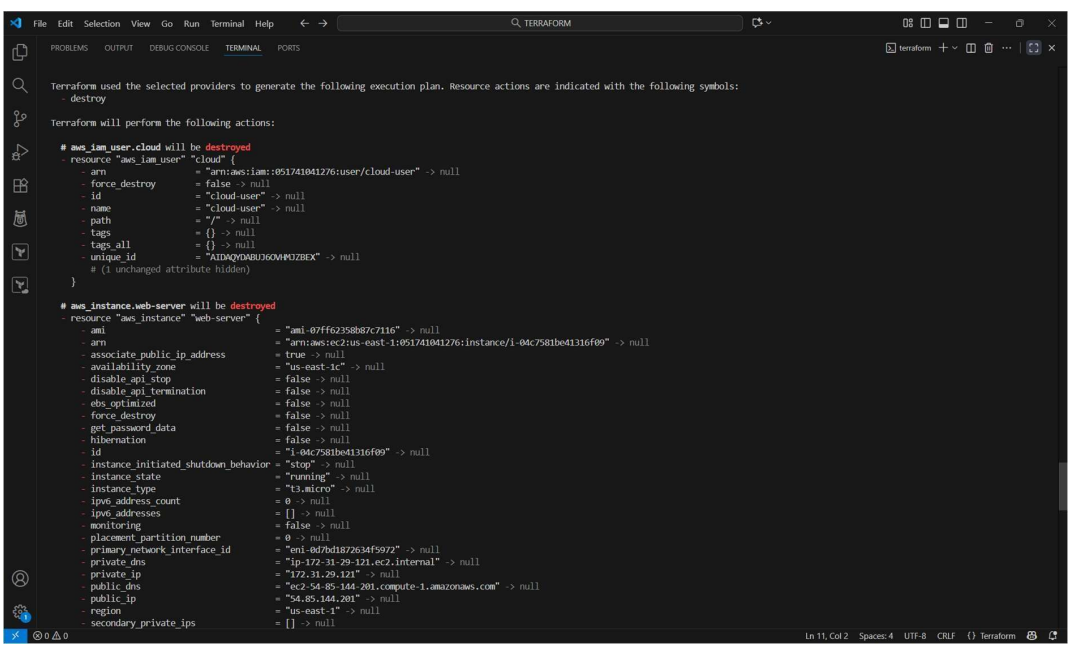

File Edit Selection View Go Run Terminal Help
TERRAFORM
EXPLORER
TERRAFORM
.terraform
.terraform.lock.hcl
main.tf
provider.tf
terraform.tfstate
terraform.tfstate.backup
OUTLINE
resource "aws_instance" "web-server"
ami
instance_type
tags
Name
TIMELINE main.tf
File Saved now
RUNNING TASKS
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
+ "Name" = "web-server"
}
+ tenancy = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
}
Plan: 1 to add, 0 to change, 0 to destroy.
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
aws_instance.web-server: Creating...
aws_instance.web-server: Still creating... [00m10s elapsed]
aws_instance.web-server: Creation complete after 17s [id=i-04c7581be41316f09]
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
PS C:\Users\hariharan\OneDrive\Documents\SLA\TERRAFORM>
Ln 2, Col 41 Spaces: 4 UTF-8 CRLF {} Terraform

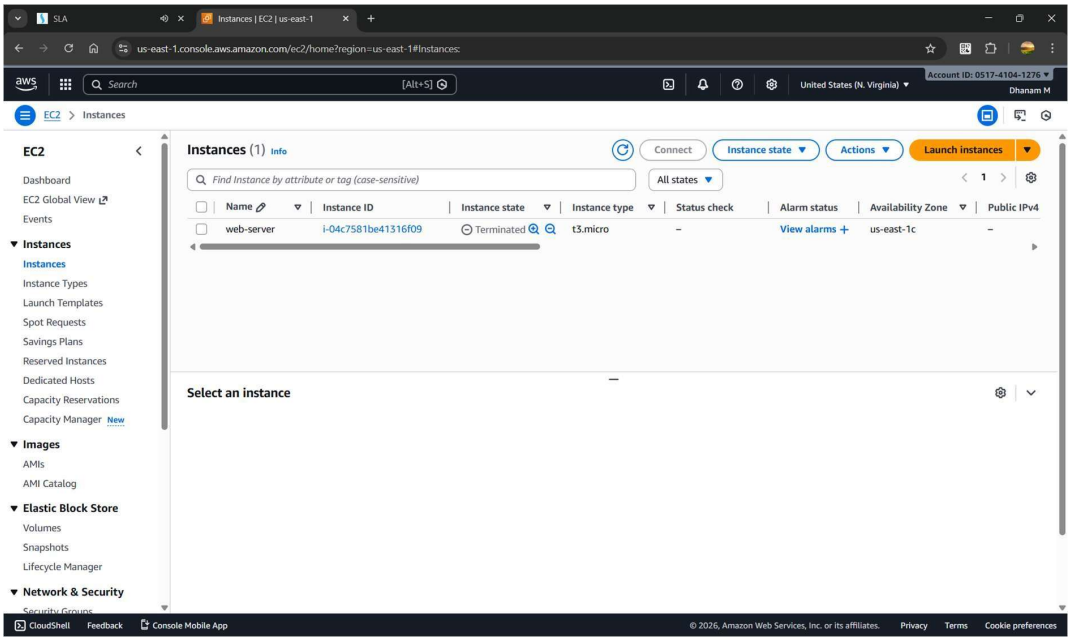

SLA
Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
aws
Search [Alt+S]
United States (N. Virginia)
Account ID: 0517-4104-1276
Dhanam M
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Instances (1) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
web-server
i-04c7581be41316f09
Running
t3.micro
Initializing
View alarms +
us-east-1c
ec2-54-85-1
Select an instance
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences



STEP:5

"TERRAFORM DESTROY –AUTO-APPROVE" COMMAND to use destroy an running instances to be terminates and delete an iam user to finish the work to be destroy it.




SLA
Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search
[Alt+S]
United States (N. Virginia)
Account ID: 0517-4104-1276
Dhanam M
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Instances (1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
web-server
i-04c7581be41316f09
Terminated
t3.micro
-
View alarms +
us-east-1c
-
Select an instance
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences